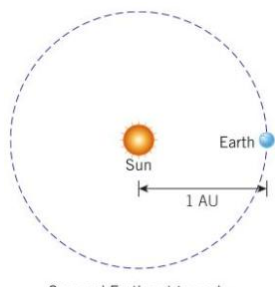
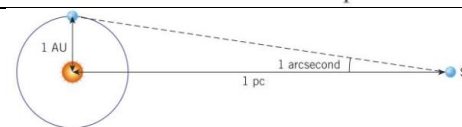


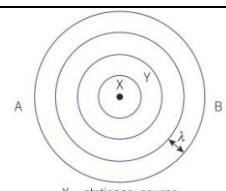
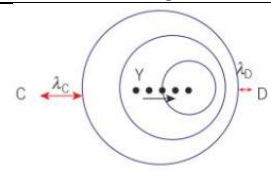
1. Lesson 1 Astronomical distances

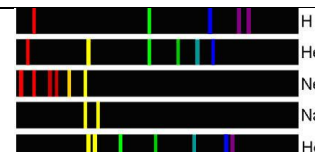
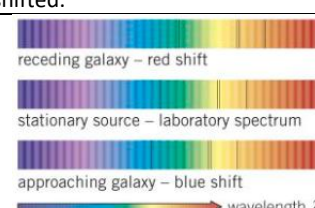
Arcminute	A minute of arc. $1^\circ = 60$ arcminutes
Arcsecond	A second of arc. 1 arcminute = 60 arcseconds $1^\circ = 3600 \text{ arcseconds}$
Cosmological principle	The assumption that, when viewed on a large enough scale, the Universe is homogeneous and isotropic, and the laws of physics are universal.
Astronomical unit (AU)	The mean distance from the Earth to the Sun, i.e. 150 million km or 1.50×10^{11} m.  Sun and Earth not to scale
Light year (ly)	The distance travelled by light in a vacuum in a time of one year (9.46×10^{15} m).
Parsec (pc)	The distance at which a radius of one AU subtends an angle of one arcsecond. $1 \text{ pc} = 3.1 \times 10^{16} \text{ m}$ $\tan(1 \text{ arcsecond}) = \frac{1 \text{ AU}}{1 \text{ pc}}$ 
Distances of some objects visible in the sky	
Object	Distance / ly
Proxima Centauri (nearest star to the Sun)	4.24
Rigel (blue supergiant star in Orion's foot)	860
Diameter of the Milky Way	100 000
Andromeda galaxy (furthest object visible with the naked eye)	2 500 000

$\theta \approx \tan \theta$	Because the angle at point S is very small, the small-angle approximation can be used. If the point S is at a distance d parsec, then the angle subtended is simply $\frac{1}{d} \text{ arcsecond}$
Stellar parallax	A technique used to determine the distance to stars that are relatively close to the Earth (less than 100 pc) by comparing their apparent positions against distant stars at times 6 months apart.
Parallax angle, p	The angle of the apparent shift in the position of a relatively close star against the backdrop of much more distant stars as the Earth makes a quarter an orbit around the Sun. $d = \frac{1}{p}$ p is measured in arcseconds d – distance to the nearby star in parsecs This technique is limited to stars less than 100 pc from the Earth, because as d increases the parallax angle decreases, eventually becoming too small to measure accurately, even with the most advanced astronomical techniques.

2. Lesson 2 The Doppler effect

Doppler effect	The change in the frequency and wavelength of waves received from an object moving relative to an observer compared with what would be observed without relative motion. The Doppler effect applies to all types of waves.
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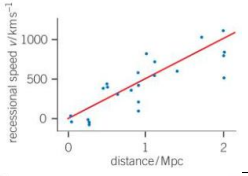
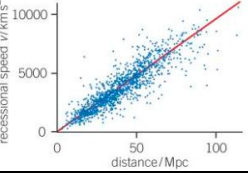
Wave source	A source of waves, such as light or sound – the object moving relative to an observer of the Doppler effect.
Stationary wave source	 X = stationary source The wave source is stationary. Both observers experience waves at the same frequency and wavelength.
Wave source is moving away	 The wave source is moving away from observer C towards observer D. For observer C the waves are stretched out, but for observer D are compressed.
Doppler shift in starlight	The Doppler effect can be used to determine the relative velocity of a distant galaxy.
Absorption line spectrum	A set of specific frequencies of electromagnetic radiation, visible as dark lines in an otherwise continuous spectrum on spectroscopy. They are absorbed by atoms as their electrons are excited between energy states by absorbing the corresponding amount of energy in the form of photons – every element has a characteristic line spectrum.

Line spectrum of different elements	
Spectral line	A line in an emission line spectrum or absorption line spectrum at a specific wavelength.
Blue-shift	The shortening of observed wavelength that occurs when a wave source is moving towards the observer – in astronomy, if a galaxy is moving towards the Earth, the absorption lines in its spectrum will be blue-shifted, that is, moved towards the blue end of the spectrum.
Red-shift	The lengthening of observed wavelength that occurs when a wave source is moving away from the observer – in astronomy, if a galaxy is moving away from the Earth (receding), the absorption lines in its spectrum will be red-shifted.
	

Doppler equation	$\frac{\Delta\lambda}{\lambda} \approx \frac{\Delta f}{f} \approx \frac{v}{c}$ λ – is the source wavelength $\Delta\lambda$ – is the change in wavelength recorded by the observer f – is the source frequency Δf – is the change in frequency recorded by the observer v – is the magnitude of the relative velocity between the source and observer c – speed of light through the vacuum
Doppler Equation limits	The Doppler equation can only be used for galaxies with speed far less than the speed of light.

3. Lesson 3 Hubble's law

Edwin Powell Hubble	American astronomer
Two key observations	1. He confirmed earlier observations that the light from the vast majority of galaxies was red-shifted, they had a relative velocity away from the Earth. 2. He found that in general the further away the galaxy was the greater the observed red shift and so the faster the galaxy was moving.
Hubble's law	The recessional speed v of a galaxy is almost directly proportional to its distance d from the Earth. $v \propto d$ $v \approx H_0 d$ The SI unit for the Hubble constant is s^{-1} Alternative units are: $kms^{-1} Mpc^{-1}$

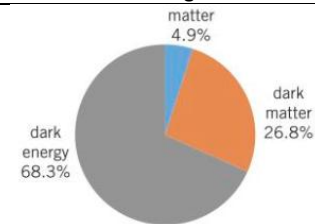
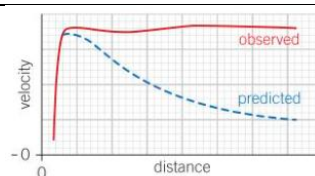
Hubble's original 1929 data	
Hubble's more recent data	
An explanation of the Hubble's law validity	The plotted data should produce a straight line through the origin. The spread in the original data points suggests that the Hubble's law is valid, but there is a large uncertainty in the value for the gradient of the best-fit line.
Hubble's constant, H_0	The gradient of a best-fit line for a plot of recessional speed against distance from Earth of other galaxies. $H_0 = 67.80 \pm 0.77 \text{ kms}^{-1} \text{ Mpc}^{-1}$ or $H_0 = 2.2 \times 10^{-18} \text{ s}^{-1}$
Key evidence	Hubble's law is the key evidence for the Big Bang theory and the model of the expanding Universe following the Big Bang.
Expanding Universe	The idea that the fabric of space and time is expanding in all directions and that as a result any point, in any part of the Universe, is moving away from every other point in the Universe, and the further the points are apart the faster their relative motion away from each other.

Cosmological principle	The assumption that, when viewed on a large enough scale, the Universe is homogeneous and isotropic, and the laws of physics are universal.
Homogeneous	Uniform in terms of the distribution of matter across the Universe when viewed on a sufficiently large scale.
Isotropic	The same in all directions (for example the Universe, appearing the same to any observer regardless of position).

4. Lesson 4 The Big Bang theory

Two key pieces of evidence for the Big Bang theory	Hubble's law - shows that the Universe is expanding and the microwave background radiation
Microwave background radiation	The microwave signal of uniform intensity detected from all directions of the sky, which fits the profile for a black body at a temperature of 2.7 K.
CMBR Detected When?	1964 – two American physicists, Robert Wilson and Arno Penzias, when they were attempting to detect signals from objects in space.
Age of Universe	Age of Universe $t \approx \frac{1}{H_0}$ $t = 4.5 \times 10^{17} \text{ s}$ (~14 billion years) We can estimate the age of the Universe by assuming that it has expanded uniformly over time since the Big Bang. Results from recent observations – the expansion of the Universe is accelerating, so this assumption is poor.

5. Lesson 5 Evolution of the Universe

Dark energy	A hypothetical form of energy that fills all the space and would explain the accelerating expansion of the Universe.
Dark matter	A Hypothetical form of matter spread throughout the galaxy that neither emits nor absorbs light – it could explain the differences between the predicted and observed velocities of stars in galaxies.
The Universe	 Current theories indicate we do not understand 95% of what makes up the Universe. 13.7 billion years old
2011 Nobel Prize in Physics	Was awarded for the discovery in 1999 that the expansion of the Universe is accelerating.
Idea of dark matter	 Observations of the velocities of stars in galaxies do not match predictions, leading to the idea of dark matter.